

SANCHARIKA DEBNATH

Data Scientist

Passionate Data Scientist adept at emerging tech for innovative solutions, with data science and back-end skills.
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WORK EXPERIENCE

Graduate Apprentice (CSE)

Defense Research and Development Organisation (DRDO), Odisha, India

JANUARY 2025 – AUGUST 2025

- Manually deployed and configured Large Language Models (LLMs) at the application-folder level in a fully offline environment for national security applications, ensuring compliance with defense protocols and strict data privacy requirements.
- Automated the generation of tender evaluation documents in $\text{\LaTeX}$ using locally hosted AI models from OLLAMA. Integrated OCR pipelines using **PyTesseract**, **EasyOCR**, and **pdfplumber** to extract text from PDFs and scanned copies, and processed content via **LangChain** for LaTeX generation (PoC). Technologies Used: **pdfPlumber**, **PyTesseract**, **EasyOCR**, **LangChain**, **Llama3.2**, **Mistral:7B**, **LaTeX**.

Product Developer

Leapon, Freelance

FEBRUARY 2024 - PRESENT

- Led RDS database and S3 Bucket migration and restructuring on AWS to support business model transitions, reducing data redundancy by **60%**. Technologies Used: **MySQL**, **AWS RDS**, **AWS S3**, **IAM**, **Django**.
- Built a blog CMS from scratch, significantly reducing the loading time by **90%** compared to the previous manual blogging system. This improved content management and site speed using: **Sanity CMS**, **REST API**, **React**.
- Implemented payment subscription system, and developed automated real-time notifications. Technologies Used: **Django**, **Stripe**, **Celery**, **Firebase**.

AI ML Engineer

Giggr Technology, Karnataka, India

JUNE 2023 - FEBRUARY 2024

- Developed an agile event system integrating facial recognition for attendance, object identification via camera data, and GPS. Achieved **87.4%** accuracy using **OpenCV**, **YOLO**, **Deep-Face**, **Geo location API**, and **JavaScript**.
- Collaborated with team to develop Neo4j architecture, integrating it collaboratively for data mining concepts, Data and Analytics using technologies such as **Neo4j**, **Graphical Database**, **REST API**, and **GCP**, **GitHub Actions**.
- Delivered a custom public API via Elasticsearch for 1.3M+ Indian education institutes, using web scraping. Technology Used: **Amazon Web Services**, **Elastic-Search**, **Shell Scripting**.
- Designed a Generative AI stack Bot based on business requirements, advanced scoring, Firebase integration, boosting interaction by **90%**. Technologies Used: **Dialogflow**, **Firebase**, **Open AI**, **NLP**, **GPT**, **Text-to-Speech**.

Back-end Developer

Leapon, Remote

FEBRUARY 2023 - JUNE 2023

- Developed a customized back-end and Django REST API for a website with 50+ advisors, integrating Unit Test and Bug fixing for feature optimization via CI/CD tools. Used **Python - Django**, **SQL**, **AWS**, **Agile Scrum**, **GIT**, **JIRA**.

Data Science Intern

Maersk Global Service Centres, Karnataka, India

JULY 2022 - FEBRUARY 2023

- Developed office seat pre-booking web app, integrating Spring Boot & Next.js, boosting user booking efficiency by **80%** in production. Technologies Used: **Spring-Boot**, **Next.js**, **Anchor UI**, **PostgreSQL**.
- Implementation of models to cluster ports using clustering & estimation methods, enhancing analytical capabilities, with **Gaussian Mixture** for Cost estimation. Utilized **Pyspark**, **Databricks**, **Microsoft Azure**.
- Approach to predict container turn time and Forecasting of Attachment-ratio in distributions logistics, considering quantitative metrics with **FLAML** as the best AutoML Utilized **ETL**, **Databricks** and **Azure ML**.

TECHNICAL SKILLS

Programming Languages: Python, C++ , R, Java, JavaScript

Technology: NLP, Machine Learning and Data Mining, Deep Learning, Computer Vision, Data Analysis, GenAI

Libraries: Pandas, NumPy, SkImage, scikit-learn, OpenCV, NLTK, spaCy, Transformers

Tools: AWS / Azure / GCP, Neo4j Graph Database, Cassandra, Pinecone, SQL, Tableau, Sanity

Frameworks: Keras, TensorFlow, PyTorch, LangChain, Llama, Django, PySpark, GitHub, Databricks, Azure ML

PROJECTS

- **Career Enchanter**: Created Generative AI data pipeline for job hunting with resume reviews, personalized recommendations, cover letter generation, job suggestions, interview preparation and ATS score calculation, using Natural Language Processing (NLP): **BERT Transformer, Gemini-Pro, Large Language Models**.
- **GenieFile**: Built an AI-powered document assistant for analyzing information across various file formats as part of the Buildspace community.. Implemented retrieval-augmented generation for efficient information retrieval and question-answering. **Llama, HuggingFace, Langchain, Vector DB, Neo4j**

PUBLICATIONS

- **Hyperspectral Image (HSI) Compression and Classification**: experimentation on a research framework for HSI using CNN bottleneck AutoEncoder to exploit **Hyper Spectral Net** for classification. Achieved **0.998** accuracy. Used **TensorFlow, Autoencoder, Hybrid SN**. Conducted **Jan'22 to Mar'22**, published in **IJCISIM**.

EDUCATION

Kalinga Institute Of Industrial Technology
Bachelor of Technology, Information Technology, CGPA: 8.83

Bhubaneswar, Orissa, India
JUNE 2019 - JULY 2023